

Meshach Ando, FE

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PROFESSIONAL SUMMARY

Civil and environmental engineering graduate with 2+ years of experience in water and wastewater design. Experienced in ArcGIS Pro, Civil 3D, Autodesk Storm and Sanitary Analysis and Python. Passed FE Civil exam and scheduled to take the PE Civil: Water Resources and Environmental exam on June 23, 2026.

EDUCATION

University of North Florida, Jacksonville, FL Aug. 2024 – May 2026
M.S., Civil Engineering

Kwame Nkrumah University of Science and Technology (KNUST), Kumasi Aug. 2018 – Nov. 2022
BSc. Civil Engineering

PROFESSIONAL EXPERIENCE

Graduate Research Assistant Aug. 2024 – May 2026
University of North Florida *Jacksonville, FL*

- Led a \$40,000 EREF-funded landfill hydrology project evaluating how stormwater infiltration and drainage network performance affect landfill cover moisture and methane oxidation rates.
- Used Python and ArcGIS Pro to analyze stormwater flow paths and runoff accumulation across landfill drainage systems.
- Conducted Monte Carlo uncertainty analyses using Civil 3D and Autodesk Storm and Sanitary Analysis to evaluate how drainage network configurations affect landfill moisture conditions and methane oxidation rates.

Quality Control Engineer Dec. 2022 – Aug. 2024
Happiness Drinking Water *Ghana*

- Monitored reverse osmosis units connected to Koyo sachet-water production lines to maintain packaged drinking water quality.
- Conducted process-control inspections on sediment filters and activated-carbon units to support potable water compliance standards.
- Evaluated treatment-line flow conditions and filtration performance to reduce production interruptions.

Staff Engineer Nov. 2022 – Nov. 2023
Kumasi Compost and Recycling Plant *Kumasi, Ghana*

- Conducted leachate flow rate measurements and effluent quality monitoring across landfill wastewater treatment operations to support effluent discharge compliance.
- Evaluated reverse osmosis treatment systems as an alternative to ultraviolet disinfection methods for improving leachate treatment efficiency and effluent quality.
- Performed laboratory-based microbial and Salmonella analyses (total coliform, fecal coliform, pathogen screening tests etc.) to verify leachate treatment performance and wastewater discharge standards.

Engineering Intern Sep. 2021 – Dec. 2021
Zoomlion Ghana Limited *Ghana*

- Inspected facultative and maturation wastewater stabilization ponds to evaluate sludge accumulation and hydraulic retention conditions within municipal waste management operations.

PROJECTS

Stormwater-Driven Methane Oxidation and Landfill Drainage Optimization Study

Aug. 2024 – May 2026

Evaluated how stormwater infiltration and drainage network performance affect landfill moisture conditions and methane oxidation using GIS analysis, hydraulic evaluations and uncertainty modeling techniques.

Whole-Landfill Methane Emissions Quantification Using Environmental Monitoring

Aug. 2024 – May 2026

Data

Developed Python- and GIS-based workflows to estimate landfill methane emissions using environmental monitoring datasets, statistical analysis and spatial modeling techniques.

Golden Hills Mixed Development Project, Abira

Jan. 2021 – May 2021

Designed drainage and sewerage discharge networks using Autodesk Storm and Sanitary Analysis for a mixed-use development infrastructure project.

CERTIFICATIONS

FE Civil – Passed

Feb. 2026

PE Civil: Water Resources and Environmental Exam Scheduled

June 23, 2026

TECHNICAL SKILLS

Hydrologic and Hydraulic: Stormwater analysis, drainage systems, wastewater treatment, landfill hydrology, flow rate monitoring

Software: Python (Advanced), ArcGIS Pro (Advanced), Microsoft Excel (Advanced), AutoCAD (Moderate), Autodesk Storm and Sanitary Analysis (Moderate), Civil 3D (Beginner), HEC-RAS (Beginner), EPA SWMM (Beginner)

Environmental and Laboratory: Effluent monitoring, leachate characterization, microbial analysis, Salmonella testing, water quality inspections

Data and Modeling: Monte Carlo simulation, uncertainty analysis, GIS analysis, environmental data analytics, statistical analysis

PRESENTATIONS

Poster Presentation – Association of Environmental Engineering and Science Professors (AEESP) Conference

Mar. 2025

Title: Environmental Systems Modeling and Landfill Emissions Research

Poster Presentation – Global Waste Management Symposium (GWMS)

Mar. 2026

Title: Modeled and Measured Landfill Methane Emissions